

C01/1/011. The coordinates axes are rotated about the origin 'O' in the counter clockwise direction through an angle of $\pi/6$. If a and b are intercepts made on the new axes by a straight line whose equation referred to the old axes is $x + y = 1$ then the value of $\frac{1}{a^2} + \frac{1}{b^2}$ is equal to :

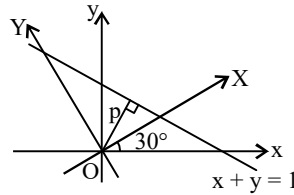
- (A) 1 (B) 2 (C) 4 (D) $\frac{1}{2}$

Ans. B

Sol. Equation of line w.r.t. new axes

$$\frac{X}{a} + \frac{Y}{b} = 1$$

$$p = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = 2$$



C02/1/012. Equation of circle touching the line $x + y = 4$ at $(1, 3)$ and intersecting the circle $x^2 + y^2 = 4$ orthogonally, is:

- (A) $x^2 + y^2 - x + 2y - 15 = 0$ (B) $x^2 + y^2 - x - y - 6 = 0$
(C) $2x^2 + 2y^2 - x + y - 22 = 0$ (D) $2x^2 + 2y^2 - x - 9y + 8 = 0$

Ans. D

Sol. Circle is $(x - 1)^2 + (y - 3)^2 + \lambda (x + y - 4) = 0$ ①

$$\Rightarrow x^2 + y^2 + x(\lambda + 2) + y(\lambda - 6) + (10 - 4\lambda) = 0$$

$$2g_1g_2 + 2f_1f_2 = 4 + \lambda_2 \Rightarrow 0 + 0 = 10 - 4\lambda - 4$$

$$\Rightarrow \lambda = \frac{3}{2} \Rightarrow x^2 + y^2 - \frac{x}{2} - \frac{9y}{2} + 4 = 0$$

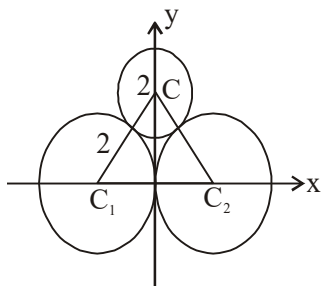
C02/1/013. Equation of a circle of radius 2 and touching the circles $x^2 + y^2 - 4|x| = 0$ is :

- (A) $x^2 + y^2 \pm 4\sqrt{3}x + 8 = 0$ (B) $x^2 + y^2 \pm 4\sqrt{3}y + 8 = 0$
(C) $x^2 + y^2 \pm 4\sqrt{3}y - 8 = 0$ (D) $x^2 + y^2 \pm 2\sqrt{3}y + 2 = 0$

Ans. B

Sol. $C(0, k)$ and $C_2(2, 0)$

$$CC_2 = 4 \Rightarrow \sqrt{4 + k^2} = 4 \Rightarrow k = \pm 2\sqrt{3}$$



C02/1/014. If tangents are drawn from any point on the line $x + y = 3$ to the circle $x^2 + y^2 = 9$ then the chord of contact passes through the point :

- (A) (1, 1) (B) (2, 2) (C) (3, 3) (D) (4, 4)

Ans. C

Sol. Chord of contact is $xh + yk = 9$

$$\Rightarrow xh + y(3 - h) = 9$$

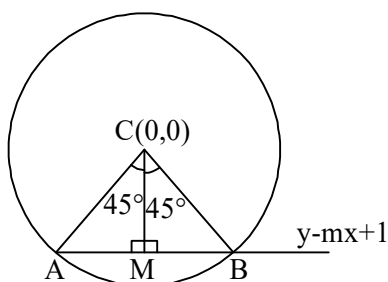
$$\Rightarrow h(x - y) + 3(y - 3) = 0 \quad \Rightarrow (3, 3)$$

C02/1/015. If the chord $y = mx + 1$ of circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle, then the value of m is :

- (A) $2 \pm \sqrt{2}$ (B) $-2 \pm \sqrt{2}$
(C) $-1 \pm \sqrt{2}$ (D) None of these

Ans. D

Sol. $CM = \frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{1+m^2}} = \frac{1}{\sqrt{2}} \quad \Rightarrow m = \pm 1$

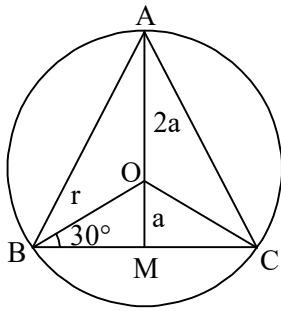


C02/1/016. The equation of circle with origin as the centre and passing through vertices of an equilateral triangle whose median is of length $3a$, is :

- (A) $x^2 + y^2 = 9a^2$ (B) $x^2 + y^2 = 16a^2$
(C) $x^2 + y^2 = 4a^2$ (D) $x^2 + y^2 = a^2$

Ans. C

Sol.



$$\sin 30^\circ = \frac{a}{r} \Rightarrow r = 2a$$

C02/1/017. If p and q be the longest distance and the shortest distance respectively of the point $(-7, 2)$ from any point (α, β) on the curve whose equation is $x^2 + y^2 - 10x - 14y - 51 = 0$ then G.M. of p and q is equal to :

(A) $2\sqrt{11}$

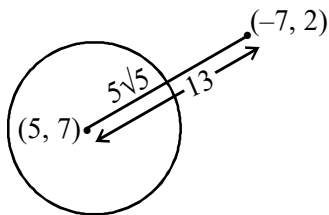
(B) $5\sqrt{5}$

(C) 13

(D) None of these

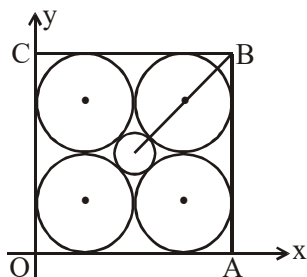
Ans. A

Sol.



$$\begin{aligned} \text{GM} &= \sqrt{(13+5\sqrt{5})(13-5\sqrt{5})} \\ &= \sqrt{169-125} = \sqrt{44} \end{aligned}$$

C02/1/018. In the figure, OABC is a square of side 6 cm, then the equation of the smallest circle is :-



(A) $(x-3)^2 + (y-3)^2 = \left(\frac{3\sqrt{2}-3}{2}\right)^2$

(B) $(x-3)^2 + (y-3)^2 = \left(\frac{3\sqrt{2}+3}{2}\right)^2$

(C) $(x - 3)^2 + (y - 3)^2 = 1$

(D) $(x - 3)^2 + (y - 3)^2 = \frac{1}{4}$

Ans. A

Sol. $r = \frac{3\sqrt{2} - 3}{2}$

$C \equiv (3, 3)$

C02/1/019. Tangents TP and TQ are drawn from a point T to circle $x^2 + y^2 = 4$. If point T lies on $3x + 4y - 12 = 0$, then the area (in sq. units) of figure formed by locus of centre of circumcircle of ΔTPQ and co-ordinate axes is :

(A) $\frac{3}{2}$

(B) 3

(C) $\frac{4}{3}$

(D) $\frac{5}{2}$

Ans. A

Sol. Let (x_1, y_1) be a point on line

$3x + 4y - 12 = 0$

Equation of circle through P and Q is

$x^2 + y^2 - 4 + \lambda(xx_1 + yy_1 - 4) = 0$

It passes through $(x_1, y_1) \therefore \lambda = -1$

Hence circle is $x^2 + y^2 - xx_1 - yy_1 = 0$

Centre $(h, k) = \left(\frac{x_1}{2}, \frac{y_1}{2} \right)$

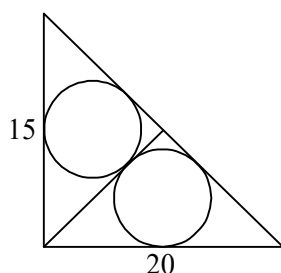
$\therefore x_1 = 2h, y_1 = 2k$

Hence locus is $3(2h) + 4(2k) = 12$

$3x + 4y = 6$

$\therefore \text{Area} = \frac{1}{2} \times 2 \times \frac{3}{2} = \frac{3}{2}$

C02/1/020. In the right angle triangle as shown, an altitude is drawn from the right angle to the hypotenuse. Circles are inscribed within each of the smaller triangles. What is the distance between the centres of these circles?



(A) 5

(B) 7

(C) 8

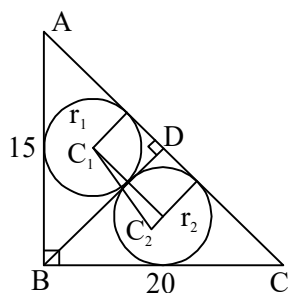
(D) $\sqrt{50}$ **Ans.** D**Sol.** Let the Δ be

$$AC = \sqrt{15^2 + 20^2} = 25$$

$$BD = 12$$

$$AD = 9$$

$$CD = 16$$



$$r_1 = \frac{\Delta}{S} = \frac{\frac{1}{2} \times 12 \times 9}{\frac{12+9+15}{2}} \Rightarrow \frac{9 \times 12}{36} = 3$$

$$r_2 = \frac{\Delta}{S} = \frac{\frac{1}{2} \times 16 \times 12}{\frac{20+16+12}{2}} \Rightarrow \frac{16 \times 12}{48} = 4$$

distance $C_1 C_2$

$$= \sqrt{(r_2 - r_1)^2 + (r_1 + r_2)^2} = \sqrt{1 + 49} = \sqrt{50}$$

C02/1/021. If the angle between the pair of straight lines formed by joining the points of intersection of $x^2 + y^2 = 4$ and $y = 3x + c$ to the origin is a right angle, then c^2 is :

(A) 20

(B) 13

(C) $\frac{1}{5}$

(D) 5

Ans. A**Sol.** $x^2 + y^2 - 4(1)^2 = 0$

$$\frac{y - 3x}{c} = 1$$

(use principle of homogenisation)